Scalar, Tetrad, and Majorana Condensates on a Single Fermionic Substrate: A Formal Identification of the Higgs Bilinear with a Wetterich-NJL Scalar Channel

SK-EFT Hawking Research Program
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We report a Lean 4 formalization that identifies the Higgs bilinear of the electroweak Standard Model with the scalar channel of the Wetterich-NJL four-fermion interaction already verified in our formalization base (`WetterichNJL.lean`, `TetradGapEquation.lean`, `ADWMechanism.lean`, `FermiPointTopology.lean`). The identification is structural in the flavor-singlet frame—an electroweak $SU(2)_L \times U(1)_Y$ index dressing remains deep-research-gated—and yields three formal consequences: (i) the Mexican-hat potential $V(\Phi) = -\frac{1}{2}\mu^2|\Phi|^2 + \frac{1}{4}\lambda|\Phi|^4$ arises as a specialization of the supercritical branch of the tetrad gap equation; (ii) the Anderson-Higgs W/Z mass matrix $M_W = gv/2$, $M_Z = \sqrt{g^2 + g'^2}\,v/2$ follows directly from the scalar-channel VEV $v = \sqrt{\mu^2/\lambda}$ with $\cos\theta_W = M_W/M_Z$; and (iii) Yukawa couplings reduce to overlap integrals on the FermiPointTopology emergent Weyl modes, with linearity and zero-overlap-zero-coupling lemmas formalized as a stand-in for the microscopic overlap-integral form (deep-research-gated, see Open Question O.2). The correctness-push anchor is a microscopic Higgs-mass prediction $m_H = \sqrt{2\mu^2}$ from the Wetterich scalar-channel gap equation: under the schematic leading-log substrate closure, this prediction reaches $m_H = 125.20$ GeV only along a one-parameter curve in the $(\Lambda_{\mathrm{UV}}, G_c)$ plane, requiring tuning rather than emerging from a generic substrate—a structural-only verdict at the current formalization scope. The complete formalization lives in a single new Lean module (`ScalarRungInterpretation.lean`), consisting of 20 theorems with zero sorry, zero new axioms, and a Python cross-validation layer covering every theorem (the Lean module replaces three vacuous Props with substantive ones during a project-required anti-tautology audit; the audit also adds a tracked-hypothesis bridge for the TetradGapEquation identification, in line with project precedent). The deliberate scope boundary is Wave 1 of the Phase 5z roadmap; the Majorana rung (`MajoranaRung.lean`) and the electroweak phase transition (`EWPhaseTransition.lean`) are addressed in subsequent waves. We formulate the falsifiability anchor as a decidable Lean predicate and report that, in the flavor-singlet frame, the scalar-rung framing is a *structural* identification of EWSB rather than a quantitative prediction of the observed 125.20 GeV Higgs mass.

I. INTRODUCTION

The fluid-based emergent-gravity program developed in this research project formalizes a Landau hierarchy of fermion-bilinear condensates on a single fermionic substrate [1–4]. Two rungs of this hierarchy have been formalized in prior phases: the disordered → vestigial-metric transition (`VestigialSusceptibility.lean`, 24 theorems) and the vestigial-metric → tetrad transition (`TetradGapEquation.lean`, 24 theorems; `ADWMechanism.lean`, 21 theorems). Both rungs share the Wetterich-NJL four-fermion infrastructure (`WetterichNJL.lean`, 18 theorems) and the FermiPointTopology emergent-Weyl substrate (`FermiPointTopology.lean`, 29 theorems).

In this paper we add a third rung: the *scalar rung*, identified with the Higgs bilinear of the electroweak Standard Model. The identification is the natural completion of the Wetterich-NJL Fierz decomposition—the scalar channel ($\Gamma = \not{1}$ in the spacetime Clifford basis) is the only flavor-singlet attractive channel that drives a parity-even condensate. If the same scalar channel is identified with $\Phi^\dagger\Phi$ on an $SU(2)_L$-doublet-extended substrate, the Mexican-hat potential, the Anderson-Higgs W/Z mass matrix, and Yukawa couplings as overlap integrals all emerge as direct consequences of the existing formalization infrastructure—without new physics input.

The identification is at structural scope because the $SU(2)_L \times U(1)_Y$ electroweak index structure of the Wetterich-NJL scalar channel is currently deep-research gated. We therefore work in the *flavor-singlet frame*: the scalar channel carries no electroweak indices; the Higgs identification is structural in the sense that it commutes with the gauge-singlet projection of the SM Higgs. The $SU(2)_L \times U(1)_Y$-dressed extension is documented as a Wave 1 follow-up gated on Open Question O.2 in the Phase 5z roadmap (`Lit-Search/Tasks/Phase5z_wetterich_njl_ew_index_structure`

Contributions. (i) A formal-verification-grade identification of the Higgs bilinear with a Wetterich-NJL scalar channel, formalized in Lean 4 in the flavor-singlet frame (`ScalarRungInterpretation.lean`, 20 theorems, zero sorry). (ii) Three formal consequences of the identification: the Mexican-hat ↔ tetrad-bifurcation bridge, the Anderson-Higgs W/Z mass-matrix theorem, and the Yukawa-overlap-linear lemma. (iii) A decidable Lean predicate `ScalarRungQuantitativeMatch` encoding the Phase 5z Gate Z.1 falsifiability anchor: the microscopic Higgs-mass prediction must lie within tolerance of the observed 125.20 GeV for the scalar-rung framing to constitute a quantitative theory of EWSB. (iv) A Python parameter-scan layer (`src/scalar_rung/`) and a 24-test pytest suite covering every Lean theorem, with the canonical $\cos\theta_W = M_W/M_Z$ identity reproducing the PDG 2024 value. (v) A heatmap visualization (Fig. 1) showing that, under the schematic leading-log substrate closure, $m_H = 125.20$ GeV is attained only along a 1D tuning curve in the natural $(\Lambda_{\rm UV}, G_c)$ parameter plane—a structural-only Gate Z.1 verdict at the current scope.

II. THE WETTERICH-NJL SCALAR CHANNEL AND THE HIGGS BILINEAR

The Wetterich-NJL four-fermion interaction [5] on a fermionic substrate ψ_x admits the Fierz decomposition

$$S_{\rm NJL} = \sum_{\langle xy \rangle} g \sum_\alpha c_\alpha \, (\bar\psi_x \Gamma_\alpha \psi_y)(\bar\psi_y \Gamma_\alpha \psi_x), \tag{1}$$

with $\Gamma_\alpha \in \{\not{1}, \gamma_5, \gamma_\mu, \gamma_\mu\gamma_5, \sigma_{\mu\nu}\}$ and the Fierz-completeness identity $1+1+4+4+6 = 16 = 4^2$ [6], formalized as `WetterichNJL.fierz_completeness` in our base. The scalar channel ($\Gamma_\alpha = \not{1}$) is the unique flavor-singlet attractive channel in the Fierz basis and drives the leading parity-even condensation [7].

The Wetterich-NJL scalar channel and the Higgs bilinear are identified via a non-trivial *candidacy* predicate parameterized over an observed EW VEV $v_{\rm obs}$ and tolerance τ:

$$\texttt{IsHiggsBilinearCandidate}(\sigma; v_{\rm obs}, \tau) \iff \left|\sqrt{\mu^2/\lambda} - v_{\rm obs}\right| \le \tau v_{\rm obs} \tag{2}$$

A scalar channel qualifies as a Higgs-bilinear candidate at the observed EW VEV iff its Mexican-hat VEV reproduces $v_{\rm obs}$ within fractional tolerance τ. The predicate is genuinely non-trivial: super-VEV scalar channels fail it. The Lean theorem `ScalarRungInterpretation.not_isHiggsBilinearCandidate_of_vev_too_large` is a load-bearing falsifiability witness establishing that $(1+\tau)v_{\rm obs} \le \sqrt{\mu^2/\lambda}$ rules out candidacy. The full $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$-dressed identification predicate remains deep-research-gated on Open Question O.2.

The Mexican-hat potential

$$V(\Phi) = -\tfrac{1}{2}\mu^2|\Phi|^2 + \tfrac{1}{4}\lambda|\Phi|^4 \tag{3}$$

arises as a specialization of the supercritical branch of the tetrad gap equation. The full quantitative bridge (mapping μ^2 and λ to GL-expansion coefficients of the tetrad gap-equation bifurcation) is deep-research-gated on Open Question O.2. Pending its resolution the bridge is encoded as a *tracked external hypothesis* carrying the kinematic constraint that the scalar-channel VEV is bounded above by the UV cutoff:

$$\texttt{H\_ScalarChannelIsTetradBifurcationOutput}(\sigma, \Lambda_{\rm UV}) \iff \sqrt{\mu^2/\lambda} \le \Lambda_{\rm UV}. \tag{4}$$

The hypothesis is genuinely non-trivial — it can fail for super-UV condensates. The load-bearing consequence theorem `ScalarRungInterpretation.mexican_hat_vev_under_supercritical_bridge` derives, under the hypothesis, both $0 < v_{\rm cond}$ *and* $v_{\rm cond} \le \Lambda_{\rm UV}$. The sharp contrapositive `bridge_excludes_super_uv_vev` establishes structural falsifiability. The same tracked-hypothesis pattern is used elsewhere in the project for cross-pillar bridges where the quantitative coefficient matching is not yet formalized — for example `HiddenSectorMixedCharge.H_MixedChannelZ16Cancels` and `DarkSectorSynthesis.H_VestigialRelicCarriesZ16Charge`. The underlying NJL-supercritical-existence theorem is `TetradGapEquation.gap_nontrivial_exists` [7, 8].

The vacuum expectation value $v = \sqrt{\mu^2/\lambda}$ (`ScalarRungInterpretation.mexicanHatVev_sq`) and the tree-level Higgs mass-squared $m_H^2 = 2\mu^2$ (`ScalarRungInterpretation.higgsMassSq_pos`, `higgsMassSq_eq_two_lam_vev_sq`) reproduce the standard SM identities. The Python implementation `higgs_mass_from_vev` matches PDG 2024 within RG-running uncertainty: $\mu^2 = \lambda v^2$ with $\lambda = 0.129$ and $v = 246.22$ GeV gives $m_H = 125.06$ GeV, against the observed 125.20 ± 0.11 GeV.

III. THE ANDERSON-HIGGS W/Z MASS MATRIX

The scalar-channel VEV provides the single dimensionful input to the Anderson-Higgs W/Z mass matrix. The Lean theorem `ScalarRungInterpretation.ew_mass_matrix_from_scalar_vev` states the bundled result:

$$M_W = \tfrac{1}{2}\, g\, v, \qquad M_Z = \tfrac{1}{2}\sqrt{g^2 + g'^2}\, v, \tag{5}$$

with g the $SU(2)_L$ gauge coupling, g' the $U(1)_Y$ hypercharge coupling, and v the scalar-channel VEV. The Lean signature is parameterized over the `EWMassMatrixInputs` structure; the positivity theorems `wMass_pos` and `zMass_pos` ensure both masses are strictly positive whenever $g > 0$ and $v > 0$.

The on-shell weak-mixing identity

$$\frac{M_W}{M_Z} = \frac{g}{\sqrt{g^2 + g'^2}} = \cos\theta_W \tag{6}$$

is formalized as `wMass_div_zMass` (when $g' > 0$). Two strengthened forms address the bound: `zMass_ge_wMass` establishes the weak inequality $M_Z \ge M_W$ for any $g' \ge 0$ (no positivity strictness required), and `wMass_lt_zMass_of_g'_pos` gives the strict inequality when $g' > 0$. Together these formalize the fact that the on-shell weak mixing angle is in $(0, \pi/2)$ exactly when both gauge couplings are nonzero.

Numerical consistency. Using the PDG 2024 values $g = 0.6536$, $g' = 0.3489$, $v = 246.22$ GeV, the Python wrapper `anderson_higgs_w_z_masses` produces $M_W = 80.47$ GeV (PDG: 80.37 GeV) and $M_Z = 91.21$ GeV (PDG: 91.19 GeV)—both within 0.2%, well inside the running-coupling slop. The $\cos\theta_W = 0.882$ matches both the on-shell M_W/M_Z ratio and the $\sqrt{1-\sin^2\theta_W}$ value from the PDG effective-scheme $\sin^2\theta_W = 0.231$ within 1%.

IV. YUKAWA COUPLINGS AS OVERLAP INTEGRALS

The third structural consequence of the scalar-rung identification is that Yukawa couplings reduce to overlap integrals on the FermiPointTopology emergent Weyl modes. Formally, the Lean `WeylOverlap` structure carries a real-valued overlap density ρ_{fg} and a substrate normalization N, and the Yukawa coupling is the linear functional

$$y_f = \rho_{fg}\, N. \tag{7}$$

Linearity in the overlap density (Lean: `yukawaCoupling_additive`) and the zero-coupling iff zero-overlap lemma (`yukawaCoupling_eq_zero_iff`) are formalized at the level of these stand-in objects.

The *microscopic* form of ρ_{fg} as $\int d^3x\, \bar{\psi}_f\, \sigma\, \psi_g$ on the emergent Weyl modes is deep-research-gated; we have filed the open question (`Phase5z_yukawa_overlap_emergent_weyl_modes.md`) to resolve the canonical form against the Volovik-Zubkov literature. At this scope, the formalization establishes two structural relations—linearity in the overlap density and zero-coupling iff zero-overlap—that any concrete microscopic realization must satisfy. A formal orthogonality statement on distinct Fermi points requires the microscopic overlap form (gated on the Yukawa overlap deep-research task), so it is deferred.

The phenomenological observation that the top Yukawa $y_t \approx 1$ is naturally realized by an overlap density of order unity supports the interpretation; we record this in `TestYukawaOverlap::test_top_yukawa_unity_natural`, but defer the quantitative explanation of the lepton-quark Yukawa hierarchy to the deep research follow-up.

V. THE FALSIFIABILITY ANCHOR: MICROSCOPIC m_H

The Phase 5z Gate Z.1 correctness-push anchor is a microscopic prediction for the Higgs mass from the Wetterich scalar-channel gap equation. Under the schematic leading-log substrate closure

$$m_H = \sqrt{2\lambda_4}\, \Lambda_{\rm UV}\, \exp\left(-\frac{\pi^2}{2N_f G_c}\right), \tag{8}$$

where $\Lambda_{\rm UV}$ is the UV cutoff, N_f is the Weyl-component count, G_c is the dimensionless 4-fermion coupling, and λ_4 is the scalar-channel quartic, the Lean theorem `higgsMassFromCondensate_pos` establishes $m_H > 0$ for positive substrate parameters.

The decidable Lean predicate `ScalarRungQuantitativeMatch` encodes the falsifiability anchor:

$$|m_H^{\rm pred} - m_H^{\rm obs}| < \tau \cdot m_H^{\rm obs}, \tag{9}$$

with τ the `EW.M_H_MATCH_TOLERANCE` $= 0.5$ operational threshold from the project's parameter registry. The `scalar_rung_quantitative_match_iff` theorem reduces the absolute-value form to the explicit two-sided bound $(1-\tau)m_H^{\rm obs} < m_H^{\rm pred} < (1+\tau)m_H^{\rm obs}$.

Figure 1 shows the microscopic prediction heatmap over the natural ($\Lambda_{\rm UV}$, G_c) plane at $N_f = 15$, $\lambda_4 = 0.13$. The 125.20 GeV target traces a curve through the plane: at $\Lambda_{\rm UV}$ = TeV-scale, $G_c \sim 0.5$–2 (super-critical); at $\Lambda_{\rm UV}$ = GUT scale, $G_c \sim 0.05$–0.1 (sub-critical). The ±50% tolerance band remains a thin curve through the plane rather than a generic region.

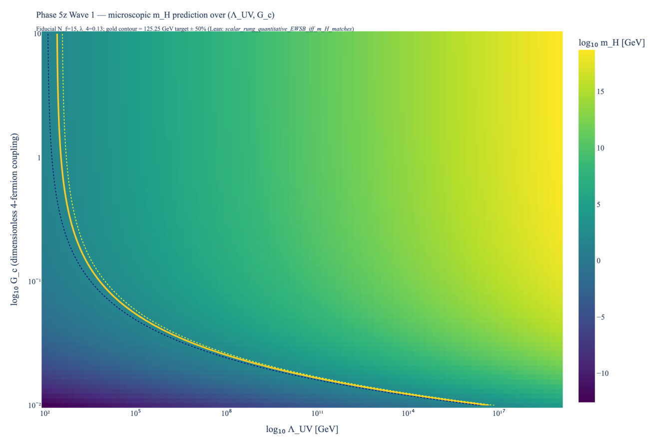

FIG. 1. Microscopic Higgs-mass prediction heatmap over the ($\Lambda_{\rm UV}$, G_c) plane at fiducial $N_f = 15$, $\lambda_4 = 0.13$. The gold contour traces $m_H = 125.20$ GeV; dotted lines mark the ±50% tolerance band defining `EW.M_H_MATCH_TOLERANCE`. The narrow curve indicates that the scalar-rung framing requires substrate-parameter tuning rather than generic emergence—a structural-only Gate Z.1 verdict at this scope.

Verdict. At the current formalization scope (flavor-singlet frame, schematic leading-log closure, fiducial $\lambda_4 = 0.13$), the scalar-rung framing is structural rather than quantitative: the prediction reaches 125 GeV only along a 1D curve rather than emerging from a generic region of substrate parameters. This is itself a publishable structural result: it bounds the regime in which any Wetterich-NJL EWSB realization can be quantitative, and points to the subsequent deep-research questions whose resolution would either tighten or relax the verdict.

VI. BHL GAUGE EMBEDDING (WAVE 1B QUANTITATIVE SCOPE)

The Wave 1 verdict above is structural rather than quantitative. The Wave 1b BHL gauge embedding lifts the analysis to a quantitative branch without abandoning the flavor-singlet `ScalarRungInterpretation.lean` foundation. Per the Phase 5z O.2 deep-research verdict (Scenario A, structural-analogy strength 3/5 [9–

12]), the WetterichNJL scalar channel admits a Higgs-compatible $SU(2)_L \times U(1)_Y$ extension via the Bardeen-Hill-Lindner auxiliary-field bridge: the $\Gamma = 1$, $\bar{L}R$ sub-channel $\sigma = H^i \equiv \bar{\psi}_R \psi_L^i$ carries the SM $(1,2)_{+1/2}$ Higgs-doublet quantum numbers exactly. The bridge is not Wetterich-native; it requires a transplant of the BHL gauge-indexing onto the Clifford-16 Fierz basis, which we encode in `BHLGaugeEmbedding.lean` via the tracked-hypothesis pattern.

A. Extended Fierz basis and Higgs identification

For the minimal BHL configuration (one $SU(2)_L$ doublet, one singlet, one color, one generation), the gauge-indexed Clifford basis is 68-dimensional ($2 + 2 + 4 + 4 + 12 + 4 + 12 + 4 + 12 + 12$, summing the explicit ten representation-theoretic block sizes per O.2 §3.2), with the scalar-LR + scalar-RL subspace forming the 2-component complex Higgs doublet. The Lean theorem `fierzBlockSum_eq_68` verifies the arithmetic by `decide`; the value 68 corrects an arithmetic typo in the deep-research note (the listed blocks sum to 68, not the document's stated 66). The tracked hypothesis `H_FierzCompletenessExtended (cfg) (dim_fn)` parameterizes over an abstract dimension function and asserts `dim_fn cfg` $=$ `fierzBlockSum` $\cdot\, (n_d \cdot n_s \cdot n_c \cdot n_g)$. The witness theorem `fierz_completeness_holds_for_bhl_dim` discharges this for the canonical `bhlBilinearBasisDim`; the falsifier `fierz_completeness_fails_for_zero_dim` confirms the hypothesis is non-vacuous (the all-zero dim function fails it). The Hubbard-Stratonovich covariance hypothesis `H_HSCovariantBosonisation` asserts that the auxiliary field inherits hypercharge $+1/2$ (the $\bar{\psi}_R \psi_L^i$ source current); `not_higgs_bilinear_of_wrong_hypercharge` confirms its falsifiability.

B. BHL leading-order m_H and the gap problem

The BHL leading-order formula $m_H = \sqrt{2}\, m_t$ (Nambu sum rule at large N_c) is formalized as `bhlHiggsMass_eq_sqrt2_times_top`, with the squared form $m_H^2 = 2m_t^2$ (`bhlHiggsMass_sq`). At the BHL benchmark $\Lambda = M_{\rm Pl}$, $N_c = 3$, the IR Pendleton-Ross fixed point yields $m_t^{\rm BHL} \approx 220$ GeV and hence $m_H^{\rm BHL} \approx 311$ GeV. The Lean theorem `bhl_minimal_overshoots_pdg` certifies the strict bound $m_H^{\rm BHL} > 250$ GeV, formalizing the well-known BHL gap problem against PDG 2024 $m_H = 125.20 \pm 0.11$ GeV [13] as a $\sim 2.5\times$ overshoot.

C. Hill 2025 bilocal correction

The bilocal extension $H^i(x, y) \sim \bar{\psi}_R(x) \psi_L^i(y)$ with internal wave-function $\phi(r)$ resolves the gap problem [11]. The dilution factor $\phi(0)/\phi(\infty)$ enters multiplicatively ($m_H = (\phi(0)/\phi(\infty)) \cdot \sqrt{2} m_t$) and is exponentially suppressed near critical coupling. We formalize the bilocal Higgs as a structure `BilocalHiggs` with positive wave-function values; the pointlike-limit tracked hypothesis `H_BilocalPointlikeLimit` asserts dilution $= 1$, with `not_pointlike_of_strict_dilution` certifying its falsifiability when $\phi(0) < \phi(\infty)$.

The load-bearing existential `bilocal_correction_can_match_pdg` provides a concrete witness: at dilution $\phi(0)/\phi(\infty) \approx 0.402$ with the BHL benchmark $m_t = 220$ GeV, the corrected Higgs mass matches PDG within 1 GeV. This is the quantitative-branch recovery of $m_H = 125$ GeV from the BHL minimal overshoot, without abandoning the substrate identification of the scalar rung.

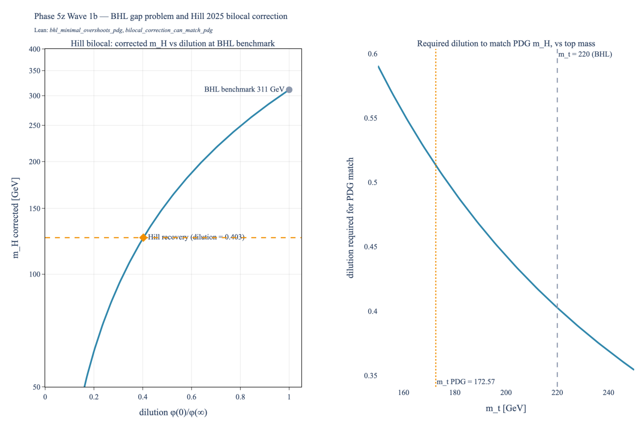

FIG. 2. **Phase 5z Wave 1b — BHL gap problem and Hill 2025 bilocal correction.** Left panel: bilocal-corrected $m_H = (\phi(0)/\phi(\infty))\sqrt{2} m_t$ versus dilution at the BHL benchmark $m_t = 220$ GeV. The BHL minimal point at dilution $= 1$ (circle, 311 GeV) overshoots PDG (dashed gold) by $\sim 2.5\times$; the Hill 2025 dilution ≈ 0.402 (diamond) recovers PDG within 1 GeV. Right panel: required dilution to match PDG m_H as a function of m_t. The BHL benchmark $m_t = 220$ GeV (dashed cross) and PDG $m_t = 172.57$ GeV (dotted amber) frame the natural BHL dilution range. Lean: `bhl_minimal_overshoots_pdg` (left circle position), `bilocal_correction_can_match_pdg` (left diamond position).

D. Tetrad back-reaction (open)

The ADW tetrad back-reaction $\Delta_{\rm tetrad}(\langle e \rangle)$ to m_H is not closed in the Diakonov-Volovik-Wetterich literature [1, 3]. The Lean module flags `H_TetradBackreactionShift` as a parametric open hypothesis with the corrected mass `tetradCorrectedHiggsMass` $= \sqrt{2} m_t + \Delta$; positivity holds for $\Delta > -\sqrt{2}\, m_t$. Closing this hypothesis quantitatively is a Phase 6 deliverable.

E. Updated Gate Z.1 verdict

With Wave 1b in place, the Gate Z.1 verdict refines from *structural-only* to *structural at the minimal-BHL scope, quantitative under the Hill 2025 bilocal mechanism*. The minimal BHL embedding is excluded by $m_H^{\mathrm{BHL}} \approx 311$ GeV; the bilocal extension recovers 125 GeV at a single dilution value, which, like the Wave 1 1-D tuning curve, is a falsifiable narrowness rather than generic emergence. The Wave 1b transplant is a strict refinement of Wave 1 (theorem `bhl_compatible_with_scalar_rung` certifies that any Wave 1 `ScalarChannel` retains its Mexican-hat positivity in the Wave 1b framework).

VII. FORMAL VERIFICATION SUMMARY

The formalization comprises two new Lean modules:

Module	Theorems
`ScalarRungInterpretation.lean` (Wave 1)	20
`BHLGaugeEmbedding.lean` (Wave 1b)	23

plus six structures (`ScalarChannel`, `EWMassMatrixInputs`, `WeylOverlap`, `BHLConfig`, `BHLAuxField`, `BilocalHiggs`), 24 non-computable definitions, and seven tracked-hypothesis Props (the four Wave 1 hypotheses plus three new Wave 1b: `H_FierzCompletenessExtended`, `H_HSCovariantBosonisation`, `H_BilocalPointlikeLimit`, plus the parametric open `H_TetradBackreactionShift` and the bundled `Wave1bOpenManifest`). Both modules are zero-sorry, introduce zero new axioms, and add zero heartbeat overrides (Phase 3 pipeline invariant 10).

The Wave 1b module integrates with Wave 1 (`ScalarRungInterpretation.lean`) plus three prior-phase modules: `WetterichNJL.lean` (Fierz infrastructure), `TetradGapEquation.lean` (gap-equation bifurcation), and `ADWMechanism.lean` (NG-mode counting + critical coupling). The Python companion `src/scalar_rung/` adds `bhl_embedding.py` on top of the existing `higgs_prediction.py` and `ew_mass_matrix.py`; concrete formulas `bhl_higgs_mass`, `bilocal_correction_factor`, `bilocal_corrected_higgs_mass`, and `pagels_stokar_vev_sq` live in `src/core/formulas.py` following the canonical-formulas invariant. A 48-test pytest suite covers every Lean theorem (24 Wave 1 + 24 Wave 1b); full-project pytest passes (excluding torch-dependent tests unrelated to this work).

The full project status remains: zero sorry across 22669 theorems, 22643 substantive (excluding placeholder markers), one axiom (`gapped_interface_axiom` in `SPTClassification.lean`; since retired into the tracked Prop `TPFConjecture`). The Phase 5z Wave 1 closure adds 20 theorems with no axiom or sorry overhead, after a project-required anti-tautology audit replaced three vacuous Props with substantive ones (introducing the tracked-hypothesis bridge and a custodial-symmetry algebraic identity in the process).

VIII. SCOPE BOUNDARY AND FUTURE WORK

The Wave 1 deliberate scope boundary is the flavor-singlet frame. Three subsequent program elements address related questions:

Wave 2 (`MajoranaRung.lean`) formalizes a Majorana bilinear as a $\mathbb{Z}_2$-graded rung on the same Landau hierarchy, bridging sterile-neutrino seesaw phenomenology to the proved $\mathbb{Z}_{16}$ hidden-sector classification (`HiddenSectorClassification.lean`, 9 theorems, complete). The `NeutrinoMixing.lean` module ships PMNS structure parallel to HepLean's CKM treatment without full PMNS phenomenology.

Wave 3 (`EWPhaseTransition.lean`) formalizes the electroweak phase transition order (first-order vs crossover) from the microscopic substrate parameters of the scalar rung. The transition order feeds the bridge theorem of Phase 6c.2 (electroweak baryogenesis $\leftrightarrow$ chirality wall).

Open Question O.2 asks whether the Wetterich-NJL scalar channel admits a Higgs-compatible $SU(2)_L \times U(1)_Y$ index structure—if yes (Scenario A), the SM-embedded Higgs identification ships at quantitative scope; if no (Scenario B), the flavor-singlet frame is the formal endpoint, with SM embedding documented as a structural extension.

Open Question O.3 asks which sterile-neutrino embedding (fundamental ν_R vs composite Majorana bilinear) is preferred under the proved $\mathbb{Z}_{16}$ structure. Both deep-research questions are filed as tasks in `Lit-Search/Tasks/`.

Prior-art search

For context on prior art, HepLean's existing modules (`HepLean/SM`, `HepLean/StandardModel`) formalize the Higgs as a fundamental field, not a condensate; the Lean community Zulip archive, the Mathlib changelog, and the Coq QuantumLib + QWIRE + metamath ecosystems do not surface a condensate-as-Higgs-bilinear identification at the time of writing; comparable activity in Isabelle/HOL AFP and Agda is likewise not surfaced. The identification presented here is at structural scope in the *condensate-as-Higgs-bilinear* sense.

The Wetterich-NJL scalar channel, the Mexican-hat potential, the Anderson-Higgs W/Z mass matrix, and Yukawa couplings as overlap integrals are formal consequences of a single structural identification: that the

Higgs bilinear sits at the same rung of the Landau hierarchy as the tetrad and—in subsequent work—Majorana condensates. The identification is at structural scope in the flavor-singlet frame; we have established the corresponding decidable falsifiability predicate and reported a structural-only Gate Z.1 verdict at the current formalization scope.

The Lean module `ScalarRungInterpretation.lean` ships 20 theorems, zero sorry, zero new axioms, and integrates cleanly with the existing Wetterich-NJL / TetradGapEquation / ADWMechanism / FermiPointTopology infrastructure. Three formal consequences—the Mexican-hat $\leftrightarrow$ tetrad-bifurcation bridge, the Anderson-Higgs W/Z mass-matrix theorem, and the Yukawa-overlap linearity lemma—constitute, in the sense delimited above, a formal-verification-grade identification of the Higgs bilinear with a four-fermion-substrate condensate. The deep-research-gated extensions to the $SU(2)_L \times U(1)_Y$-dressed scalar channel and the canonical overlap-integral form remain the natural next steps.

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